

Section A

Answer **all** questions in this section.

- 1 Collagen is the main structural protein of the various connective tissues in animals. As the main component of connective tissue, it is the most abundant protein in mammal, making up from 25% to 35% of the whole-body protein content. **Fig. 1.1** shows the structure of a collagen fibre and collage fibrils.

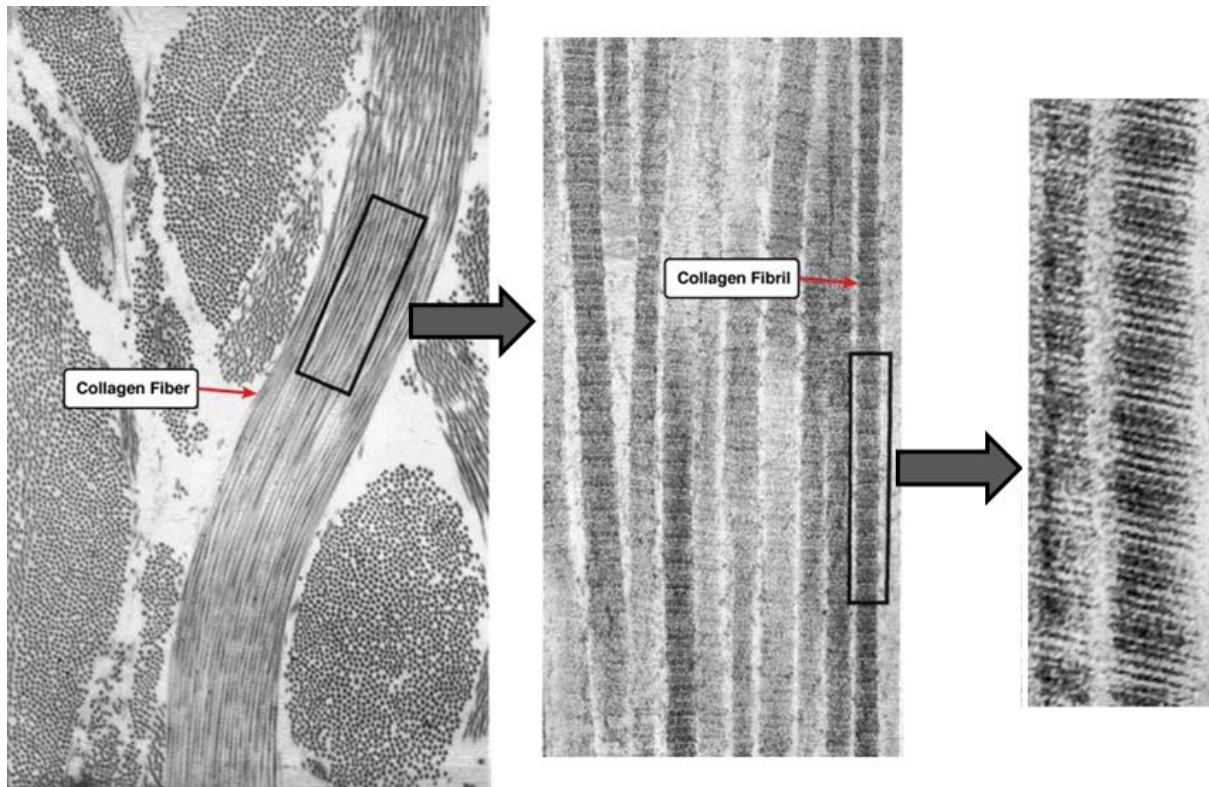


Fig 1.1

- (a) Describe the primary structure of a collagen polypeptide.

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..... [2]

(b) With reference to **Fig. 1.1**,

- (i) describe two ways how the structure of a collagen fibre differs from that of an enzyme such as amylase.

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..... [2]

- (ii) Draw a diagram to show why a collagen fibril appears as a banded structure. Annotate your diagram appropriately. [2]

(c) **Fig. 1.2** shows two membrane-bound organelles commonly found in animal cells.

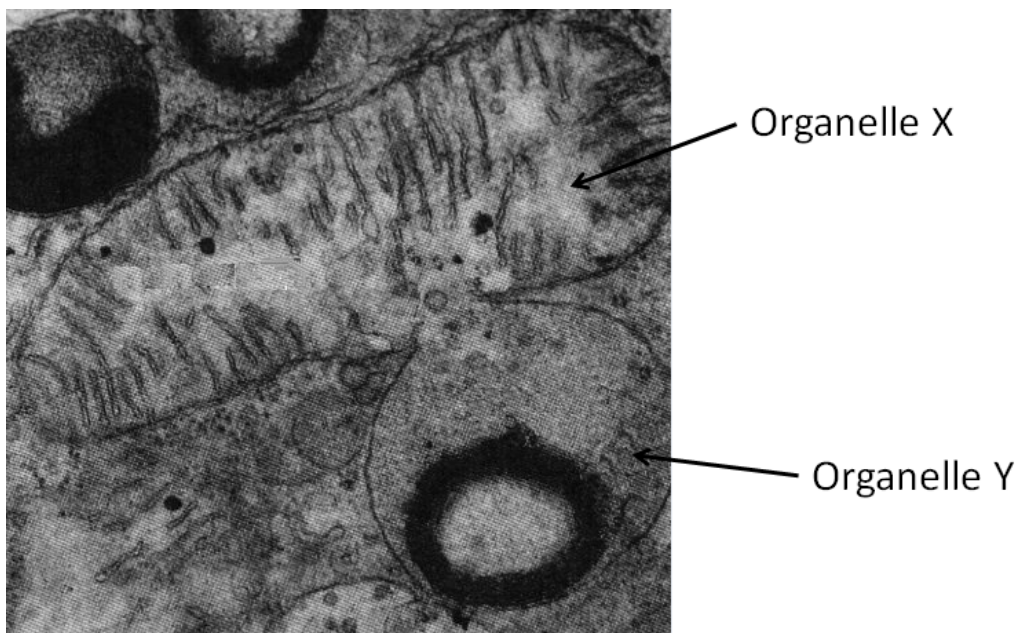


Fig 1.2

(i) State the name of the process occurring in **Fig. 1.2**.

..... [1]

(ii) Describe two other functions of organelle Y not shown in Fig. 1.2.

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..... [2]

(iii) Explain the roles of two named organelles in the formation of organelle Y.

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..... [3]

(iv) Compare two structural differences between organelle Y and a chloroplast.

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..... [2]

[Total: 14]

2 Haemoglobin S (HbS) is a variant of adult haemoglobin (HbA) that has arisen by mutation in the β globin sequence and patients with recessive HbS disorder exhibit sickle cell anaemia.

Hb A β globin	val – his – leu – thr – pro – glu – glu –lys.....
Hb S β globin	val – his – leu – thr – pro – val – glu –lys.....

Table 2.1 The first 8 amino acids of the β globin sequence of HbA and HbS

		Second base									
		U		C		A		G			
First base	U	UUU	phe	UCU	ser	UAU	tyr	UGU	cys	U	C
		UUC	phe	UCC	ser	UAC	tyr	UGC	cys		
		UUA	leu	UCA	ser	UAA	stop	UGA	stop		
		UUG	leu	UCG	ser	UAG	stop	UGG	trp		
	C	CUU	leu	CCU	pro	CAU	his	CGU	arg	U	C
		CUC	leu	CCC	pro	CAC	his	CGC	arg		
		CUA	leu	CCA	pro	CAA	gln	CGA	arg		
		CUG	leu	CCG	pro	CAG	gln	CGG	arg		
	A	AUU	ile	ACU	thr	AAU	asn	AGU	ser	U	C
		AUC	ile	ACC	thr	AAC	asn	AGC	ser		
		AUA	ile	ACA	thr	AAA	lys	AGA	arg		
		AUG	met	ACG	thr	AAG	lys	AGG	arg		
	G	GUU	val	GCU	ala	GAU	asp	GGU	gly	U	C
		GUC	val	GCC	ala	GAC	asp	GGC	gly		
		GUA	val	GCA	ala	GAA	glu	GGA	gly		
		GUG	val	GCG	ala	GAG	glu	GGG	gly		
		U		C		A		G			

Table 2.2 The genetic code (the three bases of an mRNA codon are designated here as the first, second and third bases, reading in the 5' $\rightarrow$ 3' direction along the mRNA)

(a) Use Table 2.1. and 2.2 to answer the following questions:

- (i) Explain how the change in DNA sequence of HbA β globin resulted in the changed amino acid sequence in HbS β globin and led to occurrence of sickle cell anaemia.

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..... [4]

- (ii) In another mutation in HbA, a single base substitution in the codon for the seventh amino acid occurred where the first G of the codon was replaced by U. Explain what effect this base substitution would have on β globin synthesis.

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..... [2]

- (b) While screening the blood of patients with sickle cell anaemia, the doctor found different nucleotide sequences in the chromosomal locus for haemoglobin. Given that the same Hb S haemoglobin is expressed, suggest **two** reasons for the differences noted.

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..... [2]

[Total: 8]

- 3 (a) Explain the term “retroviruses”.

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..... [2]

- (b) Explain why enveloped viruses will not exit their host cells the same way as the T4 bacteriophage.

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..... [2]

[Total: 13]

4 One causative agent of cancer is the excessive exposure to x-rays, which causes DNA damage. The tumour suppressor protein p53 plays several roles in preventing cancer. For instance, it acts as a transcription factor that up-regulates the transcription of the p21 gene, whose single product stops the cell cycle at the G1 phase, allowing time for the cell to repair the damaged DNA before it initiates DNA replication. This is shown in Figure 4.1.

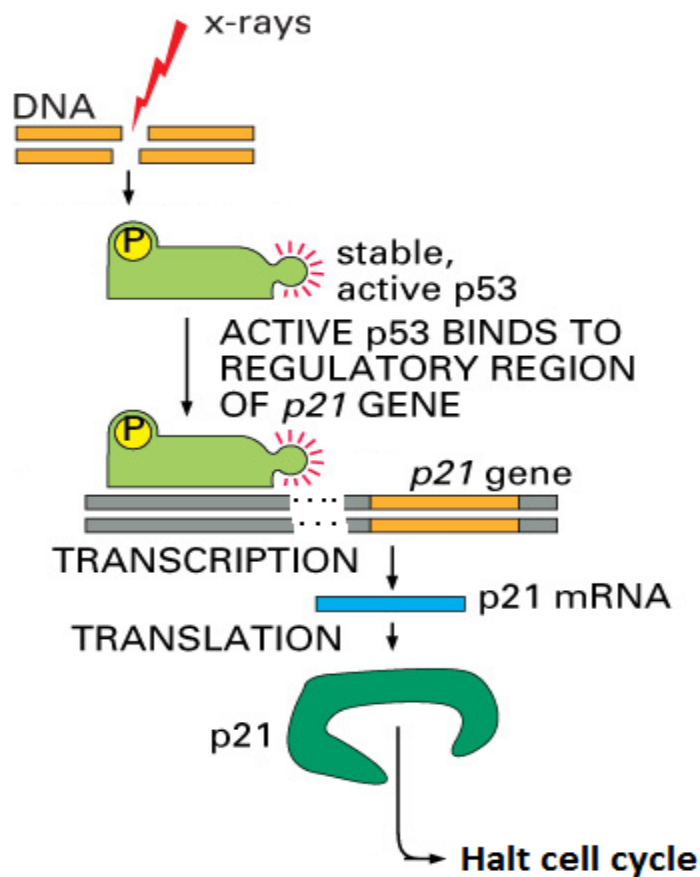


Fig 4.1

(a) (i) With reference to Fig 4.1, describe the process where the *p21* gene is transcribed into the p21 mRNA in the presence of active p53 protein.

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..... [4]

(ii) Describe how the mRNA of *E.Coli* can give rise to different protein products, unlike that of the p21 mRNA.

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..... [2]

(b) State two other ways by which p53 help prevent cancer.

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..... [2]

(c) Describe two methods of post-translational control that can be employed in a cell to control the amount of active protein in a cell.

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..... [2]

[Total: 10]

- 5 A gene found on chromosome 11 codes for an enzyme called 7-dehydrocholesterol reductase (DHCR7). This enzyme is involved in the biosynthesis of cholesterol. Inheritance of two mutated alleles results in a disease known as Smith-Lemli-Opitz syndrome, which is characterised by a reduced level of cholesterol, resulting in intellectual disabilities and physical malformation.

Another gene found on chromosome 11 codes for the protein alpha-tectorin (TECT α), which is a major component of the tectorial membrane in the inner ear. A dominant mutation results a disease known as Nonsyndromic deafness.

- (a) A man suffering from as Nonsyndromic deafness but not Smith-Lemli-Opitz syndrome was homozygous for both DHCR7 and TECT α genes. He married a woman suffering from Smith-Lemli-Opitz syndrome but not nonsyndromic deafness. They had a son who suffered only from Nonsyndromic deafness. The son married another woman with only Smith-Lemli-Opitz syndrome, who gave birth to a healthy child without either disease.
- (i) Using the symbol **A/a** to represent the TECT α gene and **D/d** to represent the DHCR7 gene, draw a genetic diagram to show how the son and his wife can give birth to a healthy child. [5]

- (ii) Explain how you could use a statistical test to show that the inheritance of TECT α and DHCR7 genes does not follow Mendel's law. No calculation is needed.

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..... [2]

- (b) Apart from DHCR7, another enzyme required in the biosynthesis of cholesterol is squalene synthase, which is coded for by a gene found on chromosome 8.

Fig. 5.1 below shows a simplified biosynthetic pathway of cholesterol as well as vitamin D. Deficiency in vitamin D results in osteomalacia, a condition in which there is softening of bones.

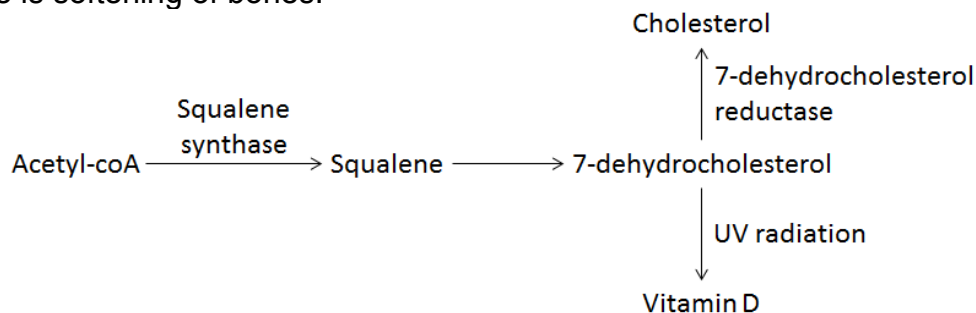


Fig. 5.1

A couple, both of them being heterozygous for both of the genes for DHCR7 and squalene synthase, exhibit a healthy phenotype with adequate level of cholesterol and vitamin D. They are expecting their first child in six months' time.

- (i) Explain the effect of independent assortment on their child's genotype.

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..... [3]

- (ii) State the probability that their first child would suffer from both Smith-Lemli-Opitz syndrome and osteomalacia.

..... [1]

[Total: 11]

6 (a) The electron micrograph Fig 6.1 shows a synapse between two neurons.

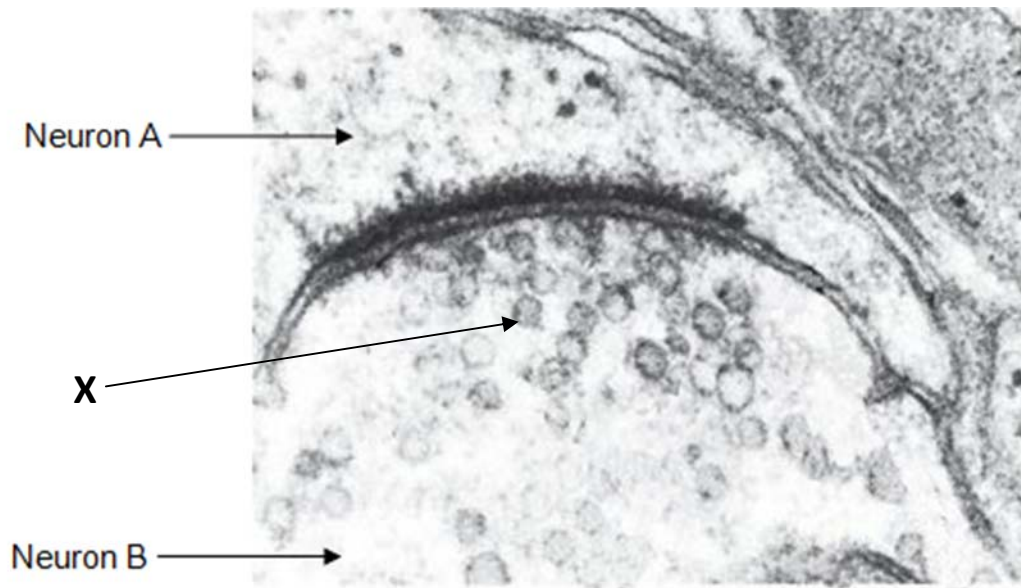


Fig 6.1

(i) Identify structure X.

..... [1]

(ii) Explain the relationship between Neuron A and Neuron B during nervous transmission.

[5]

(b) Explain the role of ATP within Neuron B in synaptic transmission.

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..... [2]

(c) Explain why transmission of impulse across a synapse is unidirectional.

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.....[2]

[Total:10]

7. The camel family, Camelidae, consists of six modern-day species that are found in three continents: Central Asia, Africa and South America. They are characterized by having only two functional toes, supported by expanded pads for walking on sand or snow. Geophysical forces such as plate tectonics and the ice-age cycles have controlled the extent of their distribution. For example, South America was separated from North America until two million years ago. The following Figure 7a shows the distribution of modern-day camels in the world.

Legend:

Recent distribution of camel

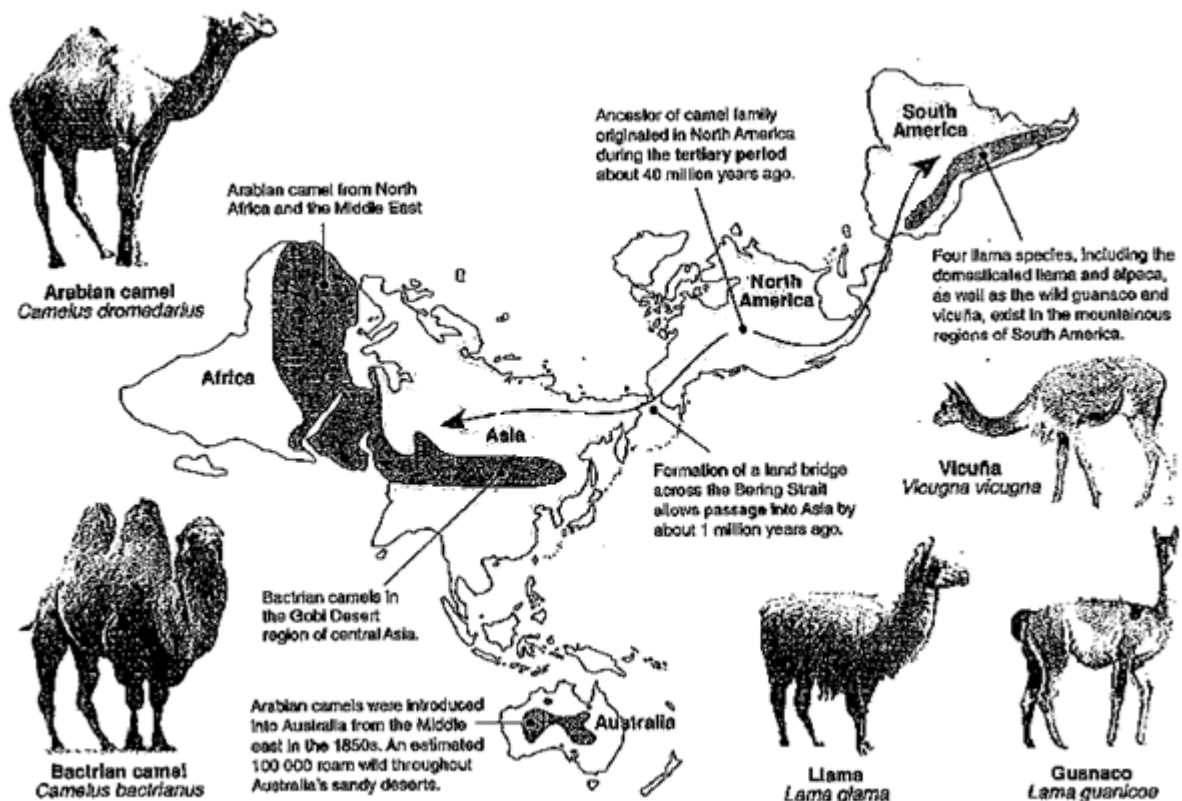


Figure 7.1: Origin and dispersal of the Camel Family

The ancestral camel *Aepyamelus* were giraffe-like and could grow to about 8 feet tall as shown in Figure 7.2.

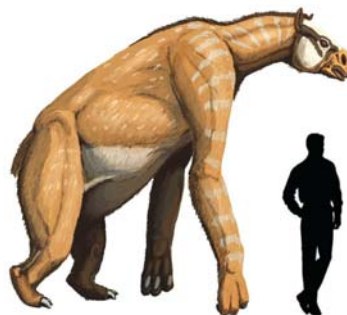


Figure 7.2: *Aepyamelus*

- (a)** With reference to Fig 7.2 and the theory of natural selection, explain the morphological differences of the llama camels (South America) and the Camelus camels (Africa) from the ancestral camel.

.....[5]

- (b)** Suggest why the South American camels have a longer evolutionary history than the African or Asian camels.

.....[1]

- (c)** Suggest why Arabian camels are found in Australia but they are no longer found in their place of origin in North America.

.....[1]

- (d) The Arabian camels in Australia typically have sand coat color. However, there is a small percentage of camels which have albino coat. It is known that the albino coat is a recessive conditions and it reduces the life-span of the camel. The population of the sand coat and albino coat camels in a region was documented.

- (i) Explain why population is considered the smallest unit of evolution.

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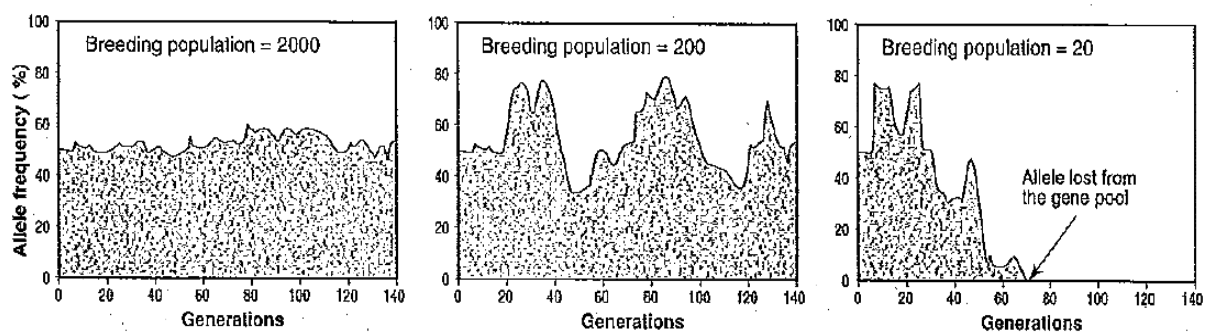
[2]

- (ii) Explain whether it is likely that the albino camel will disappear completely from the population over time. Assume there is no introduction of new albino camels into the population.

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[3]

Computer simulation of genetic drift on the coat color gene was carried out. The result shows that the allele frequency of the recessive allele in the gene pool depends on the size of the sample size of the population.



(iii) Explain how the evolutionary effect of genetic drift is affected by the sample size of the population.

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.....[2]

[Total: 14]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

Begin your answers to **(a)**, **(b)** etc. on **fresh sheets of writing paper**

- 8 (a)** Explain with examples, how environment may affect phenotype [8]
- (b)** Explain the effect of linkage on the phenotypic ratios from dihybrid crosses. [6]
- (c)** Describe how natural selection give rise to evolution. [6]
- [20]
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- 9 (a)** Explain the principles of homeostasis with reference to the regulation of blood glucose concentration by glucagon. [8]
- (b)** Explain how species are formed with respect to physiological and behavioural isolation. [6]
- (c)** Compare between ribosomes and centrioles. [6]
- [20]